



ABSTRACT

The Library Management System is a web-based application developed using Python, Flask, SQLAlchemy, and SQLite to simplify and automate library operations. Traditional library management methods often rely on manual record-keeping, which can be time-consuming, difficult to maintain, and prone to errors. This system provides a digital solution for managing books, library members, and borrowing transactions efficiently.

The application includes modules for book management, member management, and issue/return operations. It allows users to add, edit, delete, and search books, register and manage members, and issue or return books while maintaining transaction records. The system automatically calculates overdue fines at a rate of ₹2 per day after a 14-day borrowing period. A dashboard is also provided to display important statistics such as total books, total members, active borrowings, and overdue records.

The project follows a modular Flask architecture using the Application Factory Pattern and Blueprints, ensuring better code organization, scalability, and maintainability. SQLite is used as the database for storing and managing data securely and efficiently. The system also provides a user-friendly interface that makes navigation simple for librarians and administrators. By automating routine tasks, the application reduces paperwork and minimizes the chances of data duplication and human errors.

The outcome of the project is a simple, reliable, and efficient library management solution that improves record accuracy, reduces manual effort, and enhances the overall management of library resources. It also demonstrates the practical implementation of modern web development technologies in solving real-world organizational problems.

Furthermore, the Library Management System is designed to improve the accessibility and availability of library information by maintaining a centralized database of books and members. The system enables quick retrieval of records, making it easier to track book availability, borrowing history, and member activities. This not only enhances operational efficiency but also improves the overall user experience. The project serves as a practical example of how database management and web technologies can be integrated to create effective and sustainable solutions for modern library environments.

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